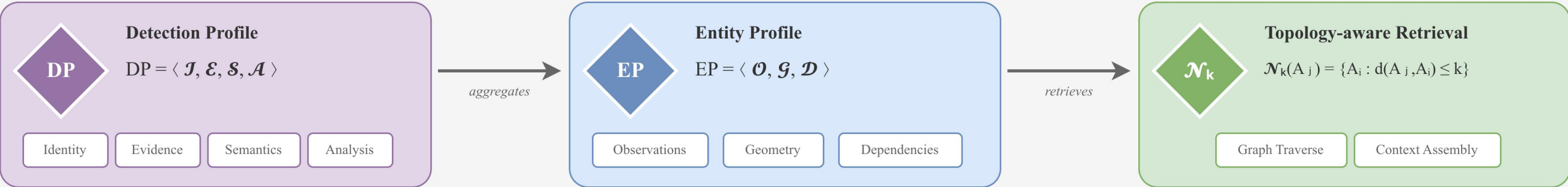


GENERIC FRAMEWORK ARCHITECTURE



Domain Instantiation

Building O&M ✓ Validated

Entity	BIM Element (Wall / Slab / Column)
DP →	IDP (Integrated Defect Profile)
EP →	IAP (Integrated Asset Profile)
Topo	Structural Load Paths (supports, adjacent_to)

Renovation Planning 🕒 Demonstrated

Entity	BIM Element + Age Metadata
DP →	Condition Assessment Record
EP →	Renovation Scope Profile
Topo	Renovation Sequence (precedes, depends_on)

Construction Progress 🏗️ Conceptual

Entity	Work Zone / Task Region
DP →	WDP (Worker Detection Profile)
EP →	WZP (Work Zone Profile)
Topo	Schedule Dependencies (successor_of, blocked_by)

Invariant: DP Structure $\langle \mathcal{I}, \mathcal{E}, \mathcal{S}, \mathcal{A} \rangle$ EP Structure $\langle \mathcal{O}, \mathcal{G}, \mathcal{D} \rangle$ k-hop Retrieval $\mathcal{N}_k(A_j)$ *Variable: Entity semantics, Edge types, Domain vocabulary*